

WIRELESS CALLER ID DISPLAY For BIKERS

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This simple telephone caller identification display can be very useful for bikers. While riding a bike or any two-wheeler, the cellphone is usually kept in a pocket. When you receive a phone call on your cellphone, you do not know who is calling unless you look at the screen. And sometimes you may not even hear the incoming ringtone. In this project, when someone calls you on your cellphone, the caller's number can be seen on the display attached to your vehicle.

Circuit and working

The circuit diagram of the wireless caller ID for bikers is shown in Fig. 1. It is built around Arduino Uno (Board1), 16x2 alphanumeric LCD (LCD1), HC-05 Bluetooth module and 12V battery. The circuit works with a 12V or 5V regulated power supply. The microcontroller (MCU) in Arduino requires 5V but the bike or motorcycle is already connected to a 12V battery.

Bluetooth module used in this project, HC-05, supports master and slave mode serial communication (9600bps - 115200bps) SPP and UART interfaces. Using these features, it can communicate with other Bluetooth-enabled devices like Android smartphones. The module runs on 3.3V to 5V power supply.

The 16x2 LCD is used to display message as soon as the bike's engine starts. Pins 3, 4, 5 and 6 of the LCD are the control lines connected to 10-kilo-ohm preset (VR1), pin 12, GND and pin 11 of Arduino, respectively. Data pins 11, 12, 13 and 14 of the LCD are

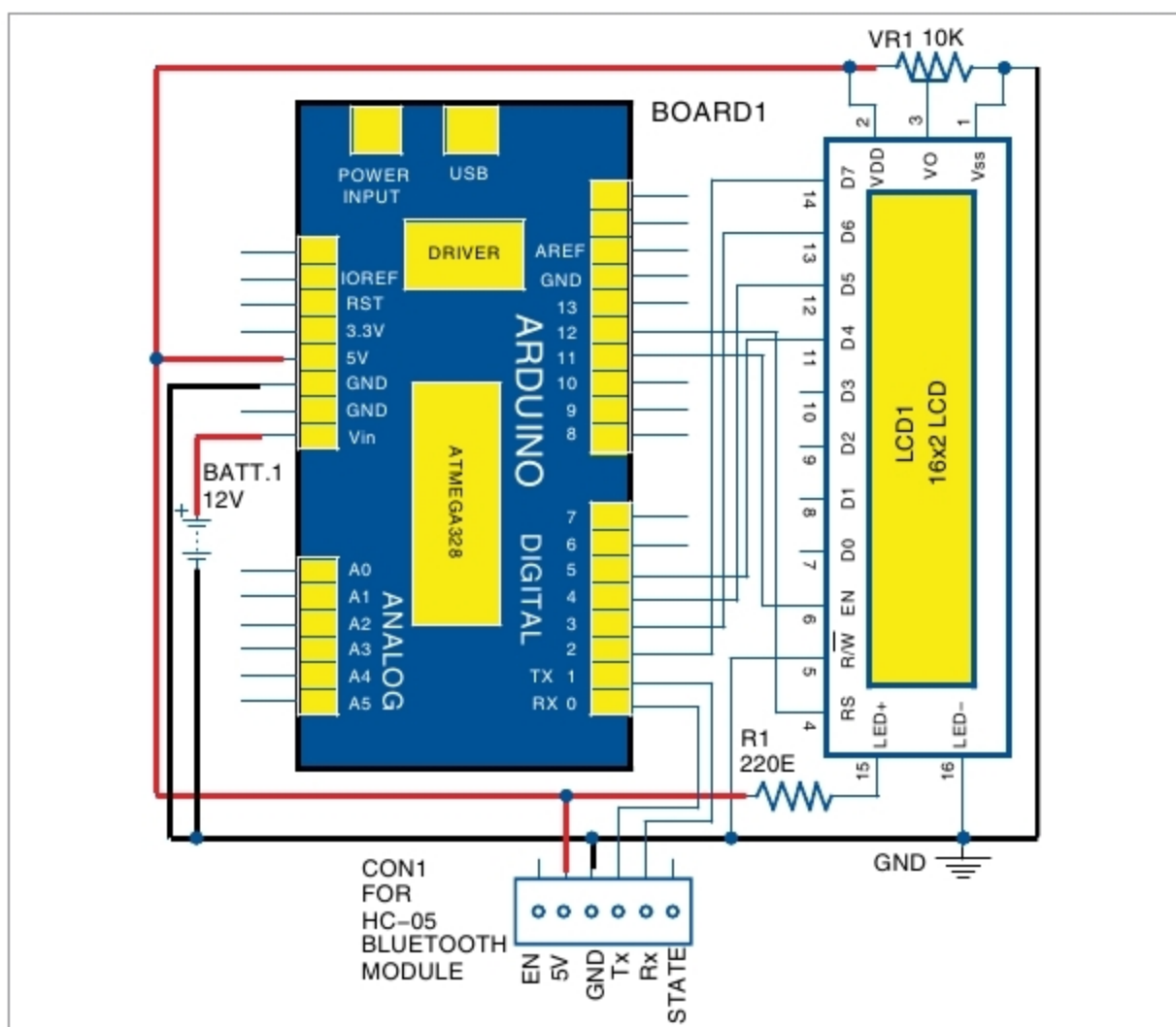


Fig. 1: Circuit diagram of wireless caller ID for bikers

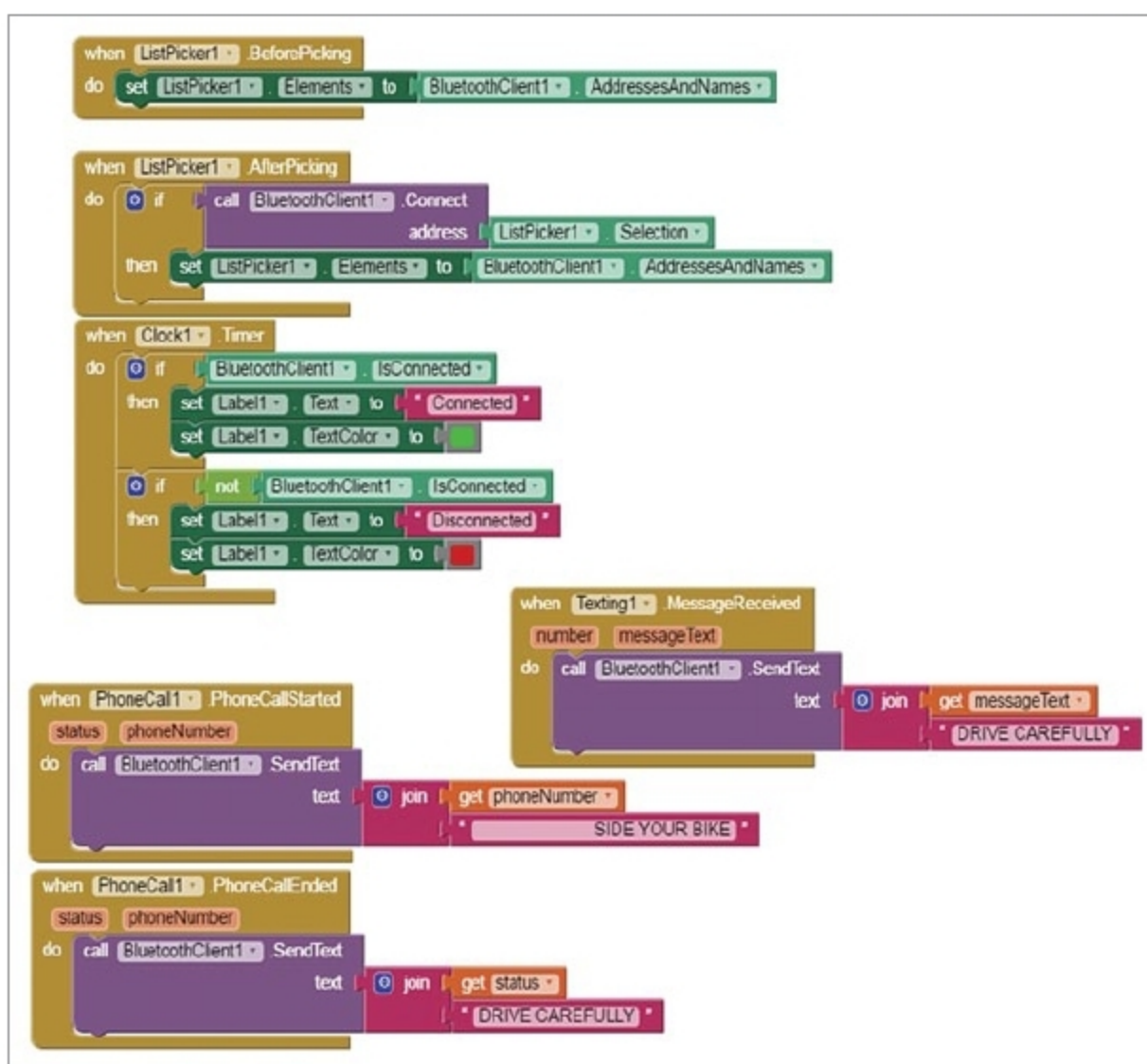


Fig. 2: Block diagram of the Android app